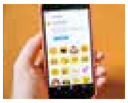
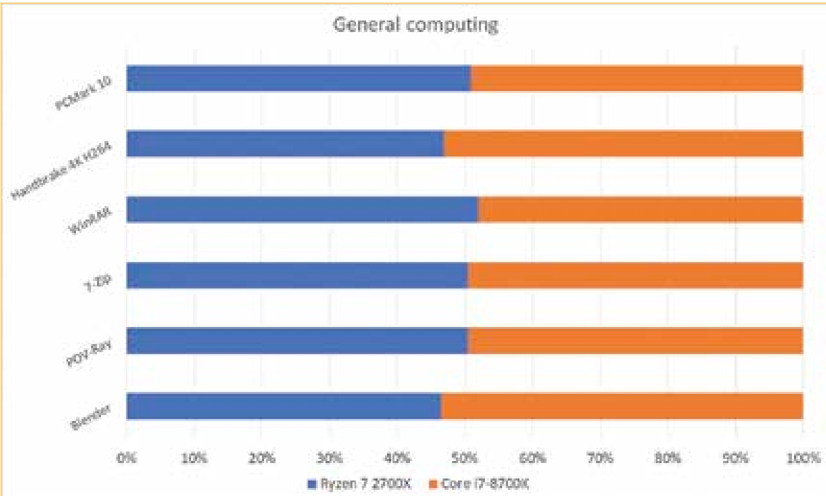
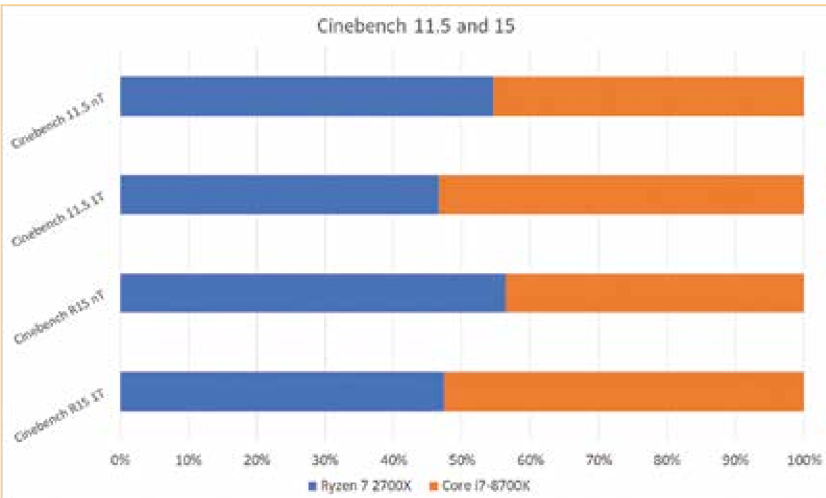


Google pauses Allo development
The head of communications at Google has decided to pause investment in Allo to focus more on Android Messages. **https://dgit.in/allopause**

Talking Vehicles coming in 2021
Lexus and Toyota will incorporate dedicated short-range communication systems in their 2021 models. **https://dgit.in/talkcar**





clocks. As for the stock frequencies, the 8700K starts off at 3.7 GHz and tops out at 4.7 GHz while the Ryzen 7 2700X has a base clock of 3.7 GHz and has a ceiling of 4.3 GHz. Overclocking the Intel CPU is ridiculously easy. Simply ramping up the multiplier takes you quite far and you only need to start tweaking the granular controls once you've hit 5.2 GHz and above. AMD's CPU, on the other hand, isn't that easy to overclock. Ryzen Master lowers the barrier significantly but you'll hit a snag before you even go past an additional 300 MHz overclock. Then onwards the dance of the clocks begins. Thankfully, scouring through HWBOT gives you a decent set of voltage values you can use to properly push the 2700X further but in comparison to Intel, it's simply not as easy. Let's see how these two CPUs perform out of the box.

POWER

Under idle conditions, the Intel Core i7-8700K consumed about 11.3 Watts which is very good. The figures were jumping between 11.2-11.4 W so we averaged it out at 11.3 W. To achieve this idle state we killed off all non-essential services, we did not install any of the bloatware that came with the motherboard and if possible we picked the lean software/driver package for any of the motherboard's components. We then ran Cadalyst Systems Benchmark and that too under the default configuration. After awhile the power draw settled down to an average of 37 W. And lastly, we ran Prime95 with AVX enabled and the total power consumption went up to 160.3 W. That's quite a lot for the Intel Core i7-8700K. We should also mention that the cooler used was a Corsair H115i PRO with the entire rig placed right in front of an air-conditioner. We did not overclock the CPU this time around.

In case of the AMD Ryzen 7 2700X, we saw the CPU consuming 12.9 W on average under the same idle conditions as the Intel CPU. However, being on a different motherboard, there's going to be a slight difference in the reported numbers. It's not an apples-to-apples comparison. Thankfully, both boards were from GIGABYTE with similar quality of components for the VRM circuitry so there shouldn't be that large of a difference in the overheads of the reported values. In the Cadalyst load test, the Ryzen 7 2700X ended up consuming lesser power than the Intel Core i7-8700K which was quite surprising. At 31 W, the difference between the two CPUs is quite significant. Then we then ran Prime95 hoping that it would go berserk. And that's exactly what the Ryzen 7 2700X did as it shot up all the way to 205 W. The delta between the two CPUs in the Prime95 test was a massive 44.7 W. So the AMD CPU has certainly improved where it matters the most – in the mid-range. After all, no PC is going to hit the ideal idle condition or be continually stressed the way Prime95 does.

GENERAL PERFORMANCE

We begin with Cinebench and it's quite evident that the AMD Ryzen 7 2700X ploughs through in the multi-threaded run whereas it still lags behind the Intel Core i7-8700K in the single-threaded run. The same can be seen in Cinebench 11.5 as well. Compared to the first gen Ryzen 7 1800X, it's a significant improvement and has come quite close to the 8700K but never crossed that of the 8700K in our benchmarks. The 8700K scored 1394 points in Cinebench 15 nT and 197 points in Cinebench 15 1T while the Ryzen 7 2700X scored 1801 and 177 in the same test. The scores scaled similarly in Cinebench 11.5 as expected.